TINGHAN (JOE) YE

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EDUCATION

Georgia Institute of Technology, Atlanta, GA
Ph.D. in Industrial Engineering *May 2028 (expected)*
Department: H. Milton Stewart School of Industrial and Systems Engineering (ISyE)

Cornell University, Ithaca, NY
B.S. (with Honors) in Operations Research and Engineering, *summa cum laude* *May 2023*
Department: School of Operations Research and Information Engineering (ORIE)

RESEARCH INTERESTS

Contextual optimization and decision-focused learning; learning-augmented optimization; sequential decision-making; agentic operations research (LLM agents for natural-language optimization modeling, algorithm discovery, and interactive decision-making); applications in transportation, logistics, supply chains, and service systems

PUBLICATIONS

* Corresponding author † Equal contribution

Journal Articles

[J5] *Contextual Stochastic Optimization for Omnichannel Multicourier Order Fulfillment under Delivery Time Uncertainty*
Tinghan Ye*, Sikai Cheng, Amira Hijazi, and Pascal Van Hentenryck
Manufacturing & Service Operations Management, 2025.
– Honorable Mention, Manufacturing & Service Operations Management Practice-based Research Competition, 2025.

[J4] *Boosting Column Generation with Graph Neural Networks for Joint Rider Trip Planning and Crew Shift Scheduling*
Jiawei Lu, Tinghan Ye*, Wenbo Chen, and Pascal Van Hentenryck
Transportation Research Part E: Logistics and Transportation Review, 2025.

[J3] *Cornell University Uses Integer Programming to Optimize Final Exam Scheduling*
Tinghan Ye*, Adam S. Jovine, Willem van Osselaer, Qihan Zhu, and David B. Shmoys
INFORMS Journal on Applied Analytics, 2025.
– Finalist, INFORMS Undergraduate Operations Research Prize, 2023.

[J2] *Managed Residential Electric Vehicle Charging Minimizes Electricity Bills while Meeting Driver and Community Preferences*
Tinghan Ye, Shanshan Liu, and Eleftheria Kontou
Transport Policy, 2024.
– 2nd Place, INFORMS Mini Poster Competition, 2021.

[J1] *A Min-Max Theorem for the Minimum Fleet-Size Problem*
Tinghan Ye and David Shmoys
Operations Research Letters, 2023.

Journal Papers Under Review

- *Learning Optimization Proxies for Sequential Contextual Stochastic Programs: An Order Fulfillment Application*
  Tinghan Ye*, Shuaicheng Tong, Changkun Guan, Beste Basciftci, and Pascal Van Hentenryck
  Major revision at **Manufacturing & Service Operations Management**.
- *Democratizing Large-Scale Re-Optimization with LLM-Guided Model Patches*
  Tinghan Ye*, Arnaud Deza, Ved Mohan, El Mehdi Er Raqabi, and Pascal Van Hentenryck
  Under review at **Management Science**.
  – Preliminary version in the **NeurIPS 2026 MLxOR Workshop**.
- *Conformal Predictive Distributions for Contextual Order Fulfillment Optimization*
  Tinghan Ye*, Amira Hijazi, and Pascal Van Hentenryck
  Under review at **Transportation Research Part E: Logistics and Transportation Review**.
  – Extended journal version of the ICCL 2025 paper [C2].

Conference Proceedings

[C6] *SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks*
Xiangyi Li, Yimin Liu, Wenbo Chen, . . . , Tinghan Ye, . . . , and Dawn Song
Advances in Neural Information Processing Systems (NeurIPS 2026).

[C5] *LISTEN to Your Preferences: An LLM Framework for Multi-Objective Selection*
Adam S. Jovine†, Tinghan Ye*,†, Francis Bahk, Jingjing Wang, Matthew Ford, David B. Shmoys, and Peter I. Frazier
International Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026).

[C4] *Plan, Verify and Fill: A Structured Parallel Decoding Approach for Diffusion Language Models*
Miao Li, Hanyang Jiang, Sikai Cheng, Hengyu Fu, Yuhang Cai, Baihe Huang, Tinghan Ye, Xuanzhou Chen, and Pascal Van Hentenryck
Conference on Empirical Methods in Natural Language Processing (EMNLP 2026).
– Spotlight, NonAR-LM Workshop at COLM 2026.

[C3] *Paratransit Optimization with Constraint Programming: A Case Study in Savannah, Georgia*
Liam Jagrowski, Kevin Dalmeijer, Tinghan Ye*, and Pascal Van Hentenryck
Principles and Practice of Constraint Programming (CP 2026).

[C2] *Conformal Predictive Distributions for Order Fulfillment Time Forecasting*
Tinghan Ye*, Amira Hijazi, and Pascal Van Hentenryck
International Conference on Computational Logistics (ICCL 2025).

[C1] *Evaluating Solvers for Linearly Constrained Simulation Optimization*
Natthawut Boonsiriphatthanajaroen, Rongyi He, Litong Liu, Tinghan Ye, and Shane G. Henderson
Winter Simulation Conference (WSC 2024).
– Code available at SimOpt.

WORKING PAPERS AND PREPRINTS

- *Decision-Focused Learning for Choice-Based Contextual Optimization*
  Tinghan Ye, Ravi Kumar, Weidong Chen, Ming Hu, and Pascal Van Hentenryck
  Working paper.
- *Decision-Focused On-Policy Learning for Contextual Linear Optimization with Partial Feedback*
  Wyame Benslimane†, Tinghan Ye†, Pascal Van Hentenryck, and Paul Grigas
  Submitted to **ICLR 2027**.
- *DecompGym: Benchmarking LLM Agents on Designing Decomposition Algorithms*
  Sikai Cheng*,†, Tinghan Ye†, Arnaud Deza, Xiaonan Qi, Miao Li, Litong Liu, and Changkun Guan
  Preprint.

- *LLM-Derived Preference Judgments Are Not Self-Consistent*
  Matthew T. Ford, Francis Bahk, Jingjing Wang, Adam S. Jovine, Tinghan Ye, David B. Shmoys, and Peter I. Frazier
  Preprint.
- *HawkesLLM: Semantic Uncertainty Propagation in Agentic Text Simulation*
  Zewei Deng, Tinghan Ye, and Liyan Xie
  ICML 2026 Workshop on Statistical Frameworks for Uncertainty in Agentic Systems.

AWARDS AND RECOGNITION

- Atlanta Air Cargo Association Ph.D. Student Fellowship for Research Excellence, Georgia Tech ISyE (2025)
- Byron W. Saunders & Allen H. Mogensen Awards for Outstanding Students, Cornell ORIE (2023)
- Cornell Engineering Learning Initiatives Undergraduate Research Award (2022, 2023)
- Tau Beta Pi Scholarship (2022)
- Research Support Grant, Illinois Office of Undergraduate Research (2021)

TEACHING

Georgia Institute of Technology

Primary Instructor

- ISYE 3103 - Introduction to Supply Chain Modeling: Logistics (B.S.): Spring 2027

Graduate Student Instructor

- ISYE 3133 - Engineering Optimization (B.S.): Fall 2026

Grader/Tutor

- Seth Bonder Camps Level 3 & 4 - Introduction to Deep Learning and Generative AI (High School): Summer 2025
- ISYE 3232 - Stochastic Manufacturing and Service Systems (B.S.): Fall 2023

Cornell University

Teaching Assistant

- ORIE 3510/5510 - Stochastic Processes (B.S. / M. Eng.): Spring 2023
- ORIE 4580/5580/5581 - Simulation Modeling and Analysis (B.S. / M. Eng.): Fall 2022
- SYSEN 5200 / ORIE 5125 - Systems Analysis Behavior and Optimization (M. Eng.): Spring 2022

University of Illinois at Urbana-Champaign

Course Assistant

- CS 125 - Introduction to Computer Science (B.S.): Fall 2020

INDUSTRY RESEARCH EXPERIENCE

Amazon, Fulfillment by Amazon (FBA) Science *May 2026 – Aug. 2026*
Research Scientist Intern *Bellevue, WA*

- Developed and benchmarked online learning policies for seller inbound arrival control using contextual bandits, reinforcement learning, and decision-focused learning, reducing the simulated backlog penalty by 10% versus the existing method
- Applied OpenEvolve to discover online learning policy implementations through LLM-guided evolutionary search

Best Buy, Fulfillment Analytics *May 2024 – May 2025*
Data Science Researcher Intern *Remote*

- Designed a stochastic optimization model for multi-item home delivery that co-optimizes fulfillment operations and delivery performance, achieving a projected 18% reduction in combined costs
- Built a machine learning model to generate probabilistic delivery-time windows, improving promise date accuracy by up to 14% and reducing late deliveries by 75%

ACADEMIC SERVICE

Reviewer

- Transportation Science
- INFORMS Journal on Computing
- IISE Transactions
- Transportation Research Part E
- INFORMS Journal on Applied Analytics
- International Journal of Production Research
- Electric Power Systems Research
- PLOS ONE
- NeurIPS 2026
- ICLR 2026
- AISTATS 2026
- INFORMS Data Science Workshop 2024–26

Session Chair

- INFORMS Workshop on Data Mining and Decision Analytics 2026
- INFORMS Workshop on Data Science 2024

CONFERENCE AND WORKSHOP PRESENTATIONS

Democratizing Large-Scale Re-Optimization with LLM-Guided Model Patches

- INFORMS Workshop on Data Mining and Decision Analytics 2026, San Francisco, CA

Learning Optimization Proxies for Sequential Contextual Stochastic Programs: An Order Fulfillment Application

- INFORMS Annual Meeting 2026, San Francisco, CA
- INFORMS Optimization Society Conference 2026, Atlanta, GA
- NeurIPS MLxOR Workshop 2025, San Diego, CA

LISTEN to Your Preferences: An LLM Framework for Multi-Objective Selection

- INFORMS Annual Meeting 2026, San Francisco, CA (poster)
- NeurIPS MLxOR Workshop 2025, San Diego, CA

Contextual Stochastic Optimization for Omnichannel Multicourier Order Fulfillment under Delivery Time Uncertainty

- INFORMS Annual Meeting 2025, Atlanta, GA
- IISE Annual Conference & Expo 2025, Atlanta, GA
- ISyE-MS&E-IOE Joint Rising Stars Workshop 2025, Atlanta, GA
- INFORMS Workshop on Data Science 2024, Seattle, WA

Boosting Column Generation with Graph Neural Networks for Joint Rider Trip Planning and Crew Shift Scheduling

- INFORMS Annual Meeting 2024, Seattle, WA

Cornell University Uses Integer Programming to Optimize Final Exam Scheduling

- INFORMS Annual Meeting 2023, Phoenix, AZ

A Min-Max Theorem for the Minimum Fleet-Size Problem

- INFORMS Annual Meeting 2022, Indianapolis, IN
- Young Researchers Workshop 2022, Ithaca, NY